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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Moon Rider
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	SmartQuote AI – WhatsApp Supplier Price Comparison & Purchase Order Drafter
Name (Leader)	Nyan Moe Htet
Email & Phone (Leader)	nyanmoehtet2001@gmail.com , +60107167645
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small businesses, including cafés, restaurants and retailers, often receive supplier quotations through messaging apps, PDFs and spreadsheets. When quotation details are scattered across formats, owners or staff must manually compare product specifications, unit sizes, minimum order quantities, delivery charges and prices. A supplier with the lowest listed price may not offer the lowest total cost, and missing or inconsistent information can lead to purchasing mistakes, wasted staff time, or avoidable expenses.

Smart Quote AI will focus on this specific task. It will turn sample supplier messages and quotations into structured records, compare only products that meet the buyer's requirements, calculate estimated order costs, and flag missing details or mismatched units for human review. The system will create a purchase order draft, but it will not send it automatically. An authorized person must review and approve it before it is sent or processed.

The problem matters because small businesses have limited time and resources to manage purchasing efficiently. Improving procurement accuracy can support SME productivity and competitiveness. Our team will validate the assumed workflow with SME owners or staff before making claims about how widespread the problem is. Initial testing will use synthetic supplier data, not real customer or company records.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Smart Quote AI is a web application that helps Malaysian SMEs compare supplier quotations and prepare purchase orders. A user can paste sample WhatsApp-style messages or enter quotation details from a PDF or spreadsheet. The prototype will extract fields such as supplier name, product, pack size, unit price, minimum order quantity, delivery fee, and quotation validity. The system will validate the fields and flag missing details or non-equivalent product units instead of making an unsupported comparison.

A deterministic Python rules engine will calculate comparable order totals and rank eligible quotations against the buyer's requested quantity and requirements. An AI/GenAI API may assist with extracting unstructured text and explaining differences in plain language; arithmetic and eligibility checks will remain code-based. The application will show the reasons behind its recommendation and allow a user to approve, reject or choose another supplier. After approval, it will generate a purchase-order draft and record the action.

We plan to use skills from the SDC workshops, including GenAI API integration, database design and cloud deployment, subject to the tools actually covered and available to our team. The MVP will use synthetic data and simulated message input; live WhatsApp integration and real purchasing transactions are outside the initial scope.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies and Frameworks

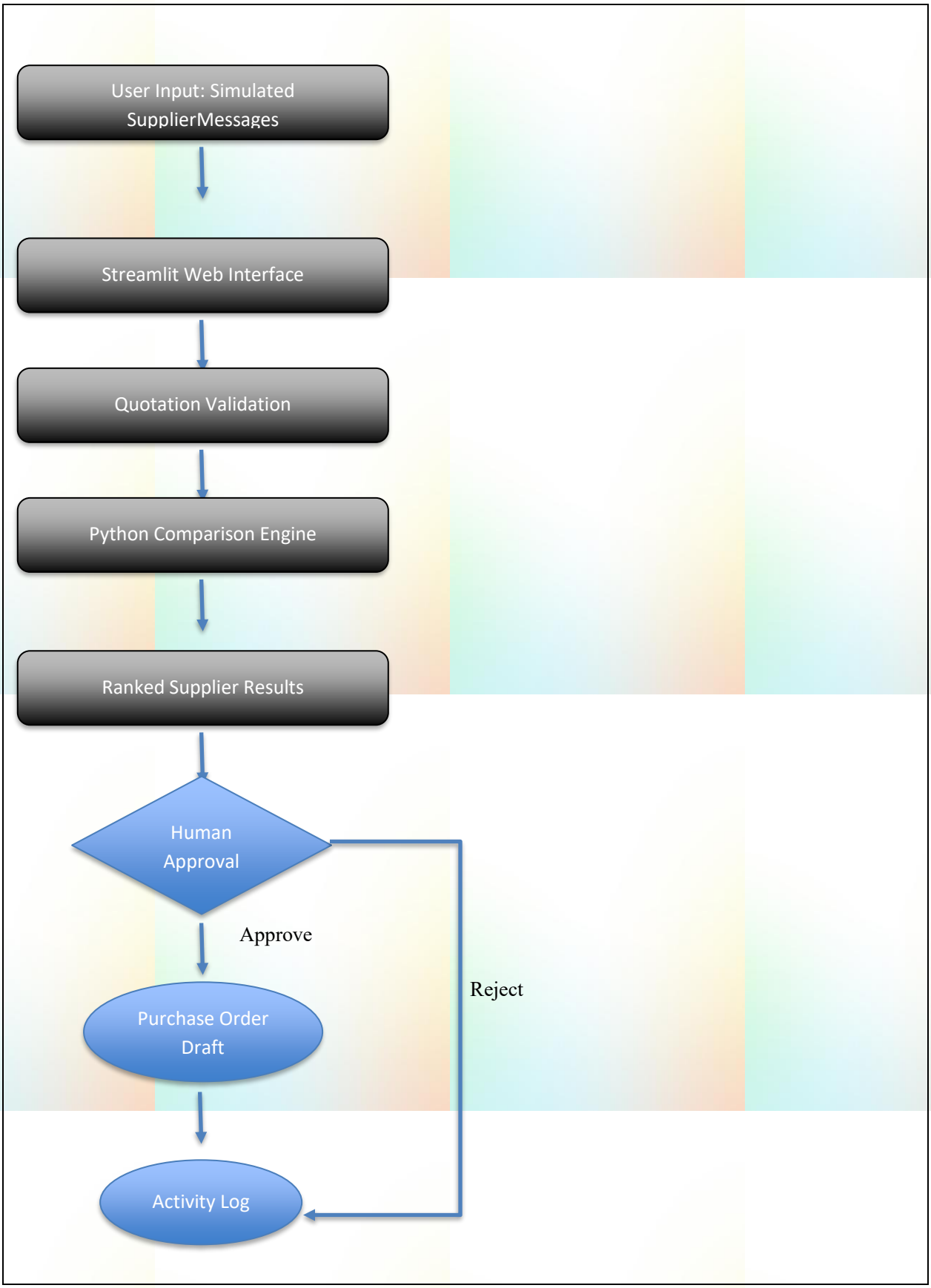
- **Programming language:** Python.
- **Web framework:** Streamlit for the user interface.
- **Application modules:** app.py, quotations.py, comparison.py, and purchase_workflow.py.
- **Decision logic:** Python-based validation, minimum order quantity checks, delivery-fee calculations, total-cost comparison, and supplier ranking.
- **Data storage:** Session-based application state and synthetic quotation data for the current prototype. A persistent database may be considered for future development.
- **Development and deployment:** Replit for development and hosting, and GitHub for source-code version control.
- **GenAI API:** AI-assisted quotation extraction and plain-language explanations are planned enhancements. They are not claimed as implemented features in the current prototype.

System Architecture

The user enters simulated supplier messages or quotation details into the web application. The application validates the information and passes eligible quotation data to the Python comparison engine, which calculates total costs and ranks suppliers. The user reviews the results and approves or rejects the proposed supplier. Following approval, the system generates a purchase-order draft and records the action in the activity log. The prototype does not automatically send purchase orders or execute transactions.

Official SDC Sponsor-Provided API or Cloud Platform

The current prototype uses Replit for development and hosting and GitHub for version control. No official SDC sponsor-provided GenAI API or cloud platform is claimed as integrated at this stage. Any future sponsor-provided integration will be identified once it has been confirmed and implemented.


User Input: Simulated SupplierMessages
Streamlit Web Interface
Quotation Validation
Python Comparison Engine
Ranked Supplier Results
Human Approval
Approve
Reject
Purchase Order Draft
Activity Log

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

SmartQuote AI follows an iterative, user-centred development approach. The team identified supplier quotation comparison as a potential administrative challenge for Malaysian micro and small businesses, particularly when prices, pack sizes, minimum order quantities and delivery fees are presented in different formats.

The team translated this problem into a focused MVP with quotation input, field validation, like-for-like comparison, transparent cost calculations, human approval, purchase-order drafting and an activity log. The current prototype uses synthetic supplier data to test the workflow without exposing real business information.

Testing focuses on whether the system calculates total costs correctly, excludes ineligible offers, flags missing or incompatible information, prevents rejected offers from generating purchase-order drafts, and preserves human control over purchasing decisions.

Before Phase 2, the team plans to gather feedback from SME owners or staff, refine the interface based on usability findings, expand test cases, and measure task completion time against a documented manual baseline. Further deployment work will assess data persistence, access controls, privacy safeguards and any required AI API integration. These validation and deployment activities remain planned until completed.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8, Target 8.2: Achieve higher levels of economic productivity through diversification, technological upgrading and innovation. Target 8.3: Promote productive activities, entrepreneurship, innovation, and the growth of micro-, small- and medium-sized enterprises. (Source: United Nations, Goal 8 — https://sdgs.un.org/goals/goal8)
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	15 minutes per supplier-quotation comparison task, used as a preliminary planning estimate rather than a verified industry statistic. The team will measure the time required to compare a fixed set of synthetic quotations manually before evaluating the prototype.

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in median time (minutes) required to compare a fixed set of supplier quotations and identify an eligible offer. Manual and prototype workflows will use the same quotation data and purchasing requirements.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a 40% reduction in median quotation-comparison time against the measured manual baseline by the final prototype presentation. If the initial 15-minute estimate is confirmed, this would mean completing the task in 9 minutes or less. This is a target to be tested, not an achieved result.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [ ] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✓ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type**(code/dataset/model)	**License**
Streamlit	Open-source web framework	Apache-2.0
Other Python packages actually used	Software libraries	Verify each package's license
Synthetic supplier quotation examples created by your team	Dataset	Team-created

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

Nyan

[Date]

24.10.2026